

ASSIGNMENT #1

Course: **CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS**
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: _____
Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100
Your score.....

TABLE #1

Symbol	Definition, Units and Data
T	Temperature, K
P	Pressure, bar
V	Molar volume, m ³ /mol
R	Universal Gas Constant, J/(mol · K)
C_v	Heat Capacity under constant volume, J/(mol · K)
S	Molar Entropy, J/(mol · K)
U	Molar Internal Energy, J/mol
Symbol	Data for Problem 1B
C_p	Heat Capacity under constant pressure, cal/(mol · K)
T	Temperature, K
A	7.953090666
B	19.09166906
C	2086.9
D	9.936466991
E	991.96

Note: All problems are worth 100 points. Your final score will be the average value

Problem 1A: Review of Differential and Integral Calculus in Thermodynamics

Prove the following identity. Hint: Start with the total differential of $\{U = U(T, V)\}$

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T \left(\frac{\partial^2 P}{\partial T^2}\right)_V$$

Problem 1B: Review of Differential and Integral Calculus in Thermodynamics.

An ideal gas is undergoing a reversible isentropic expansion from $\{T_1 = 0^\circ\text{C}, P_1 = 30\text{ bar}\}$ to $\{T_2, P_2 = 1\text{ bar}\}$. Calculate the temperature T_2 at the end of the expansion process using the secant algorithm. Use the following initial guesses for the temperature to start the algorithm: $\{T_2^{(0)} = -100.0^\circ\text{C}, T_2^{(1)} = -120.0^\circ\text{C}\}$

The differential equation that describes the isentropic expansion is

$$dS = C_p \frac{dT}{T} - R \frac{dP}{P}$$
$$C_p = A + B \frac{\left(\frac{C}{T}\right)^2}{\left(\sinh\left(\frac{C}{T}\right)\right)^2} + D \frac{\left(\frac{E}{T}\right)^2}{\left(\cosh\left(\frac{E}{T}\right)\right)^2}$$

Problem 1C: Review of Differential and Integral Calculus in Thermodynamics.

Derive the ideal gas law $\{PV = RT\}$ from the Sackur–Tetrode equation for the entropy of a monatomic ideal gas. Hint: Start with the total differential of $\{U = U(S, V)\}$